

ELECTRAL MACHINES:-

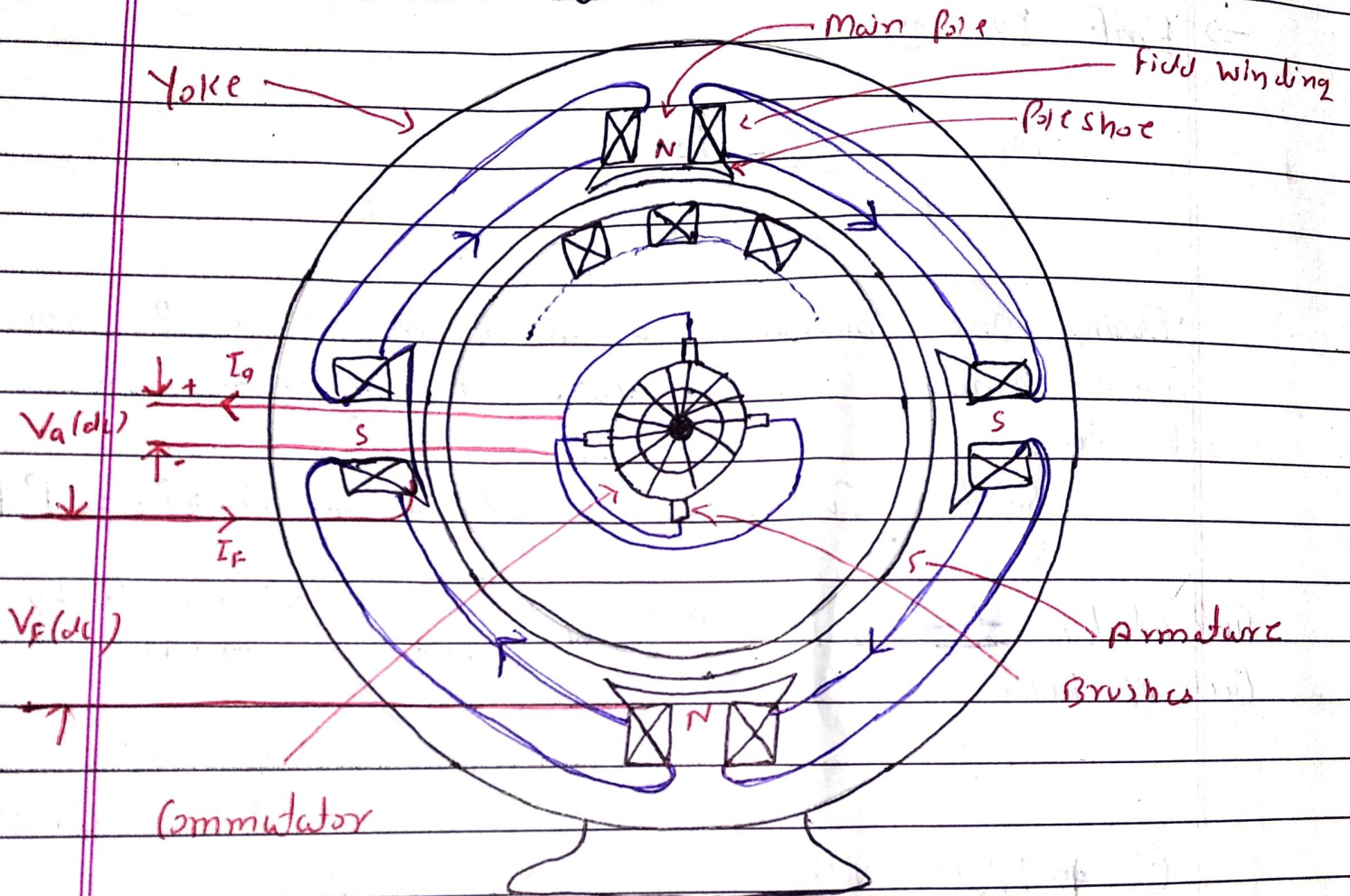
UNIT-I

- D.C machines:- Constructional Features, Principle of operation, EMF eqⁿ, Voltage build up phenomenon in DC shunt generator, characteristics of diff types of generators, Principle of operation of DC motor, back emf, Speed and Torque eqⁿ, Various characteristics of diff motor, starters and speed control of DC motor, App of DC generators and motors.

ELECTRICAL MACHINES

UNIT-I

D.C. MACHINES



4-Pole DC Machine

⇒ Operation:-

The Commutator is a Form of Rotating switch placed b/w the armature and external circuit and so arranged that it reverses the connections to external circuit at the instant of each reversal of the current in the armature coil, and thus converts induced A.C. in armature coils into DC in the external load circuit.

- In case of DC motor, DC passes through the brushes and commutator to the armature winding, while it passes through the commutator it is converted into AC so that the group of conductor's under successive field poles carry currents in opposite direction

⇒ Emf Equation

As,
$$e = - \frac{d\phi}{dt}$$

- During one revolution of armature in a pole generator each armature conductor cuts the magnetic flux ϕ times,

So flux cut by one conductor in one revolution = $P\phi$

Flux cut by ~~ϕ~~ $\phi P \times \left(\frac{N}{60} \right)$ w
Each conductor

No of revolution made by armature per minute

To convert in second.

No of Revolution of armature per second.

$$e = \frac{\phi P N}{60} \text{ volt}$$

av. emf

$$E = \frac{\phi P N}{60} \times \left(\frac{Z}{A} \right) \text{ volt}$$

no of armature conductor's per parallel path

Total emf generated
blw terminals

Let

$$\frac{P \cdot Z}{60 A} = K \Rightarrow$$

$$E = K \phi N$$

$$E \propto \phi N$$

$$E \propto \phi \left(\frac{2\pi N}{60} \right)$$

Angular velocity
in radian/second

$$\left(\frac{2\pi N}{60} \right)$$

- The Polarity of induced emf depends upon the direction of M.F and direction of Rotation, if either of two is Reversed the polarity of induced emf either brushes reversed.

~~Q. A dynamo has a rated armature current of 250 A, which is the current per path of the armature if the armature winding is simplex wave wound or simplex lap wound? The machine has 12 poles.~~

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Given, Rated current = 250 A = I_r

No of poles, $P = 12$

~~No of~~

With Simplex wave winding, No of Parallel Paths, $A = 2$

$$\text{Current per path, } I = \frac{I_r}{A} = \frac{250}{2} = 125 \text{ A}$$

With Simplex lap winding,

No of Parallel Paths, $A = P = 12$

$$\text{Current per path, } I = \frac{I_r}{A} = \frac{250}{12} = 20.83 \text{ A}$$

Q. A dc generator has an armature emf of 100 V when the useful flux per pole is 20 mWb, and the speed is 800 rpm. Calculate the generated emf (i) with the same flux and speed of 1000 rpm, (ii) with a flux per pole of 24 mWb and speed of 900 rpm.

Ans:-

Given,

$$E_1, \text{ Generated emf} = 100 \text{ V}$$

$$\Phi_1 = 20 \times 10^{-3} \text{ wb}$$

$$N_1 = 800 \text{ rpm}$$

$$[E \propto \Phi N]$$

(i)

$$\Phi_2 = \Phi_1$$

$$N_2 = 1000 \text{ rpm}$$

$$\frac{E_2}{E_1} = \frac{N_2 \Phi_2}{N_1 \Phi_1} = \frac{1000 \times 20 \times 10^{-3}}{800 \times 20 \times 10^{-3}}$$

$$E_2 = \frac{10}{8} \times 100 = \underline{\underline{125 \text{ V}}}$$

(ii)

$$\Phi_3 = 24 \times 10^{-3} \text{ wb}$$

$$N_3 = 900 \text{ rpm}$$

$$\frac{E_3}{E_2} = \frac{\Phi_3 N_3}{\Phi_2 N_2} = \frac{24 \times 10^{-3} \times 900}{20 \times 10^{-3} \times 1000}$$

$$E_3 = 1.08 \times 125 = \underline{\underline{135 \text{ V}}}$$

Q1:-

Calculate the voltage induced in the armature winding of a 4 pole, lap wound dc machine having 728 active conductors and running at 1800 rpm, the flux per pole is 30 mwb?

Given,

$$\Phi = 30 \times 10^{-3} \text{ wb}$$

For Lap Winding

$$A = P = \cancel{728} \rightarrow 4$$

(Pole = 4)

$$N = 1800 \text{ rpm}$$

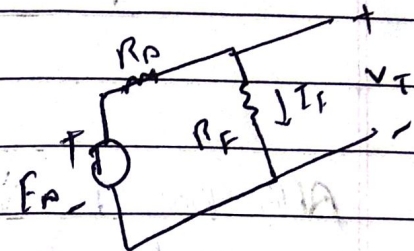
$$Z = 728$$

$$E = \frac{\Phi Z N P}{60 \times A}$$

$$= \frac{30 \times 10^{-3} \times 728 \times 1800 \times 4}{60 \times 4}$$

$$E = 655.2 \text{ V}$$

⇒ Voltage build up phenomenon



if shunt generator is producing current for its own field, so how does it initially get's the voltage??

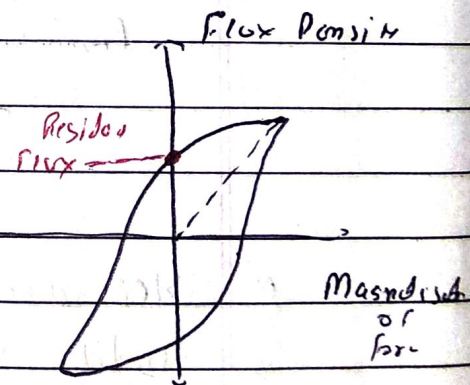
- It is generated (voltage) is because presence of residual flux in pole of ~~machine~~ generator.

↓ the magnetic flux density that remains in the material when magnetization force is zero.

~~Self generated~~
Residual EMF

Residual flux

$$E_{g, \text{residual}} = K \phi \omega \approx (1 \text{ V or } 2 \text{ V})$$



$$I_f = \frac{E_{g, \text{residual}}}{R_f} \rightarrow (\text{initially } I_f = 0, \text{ but as } E_g \text{ generated } I_f \text{ increases})$$

as I_f increases ϕ_{residual} increases

$$\uparrow I_f = \frac{E_g}{R_f} \rightarrow \phi \uparrow \Rightarrow$$

$$\rightarrow E_g \uparrow = K(\phi \uparrow) \omega$$

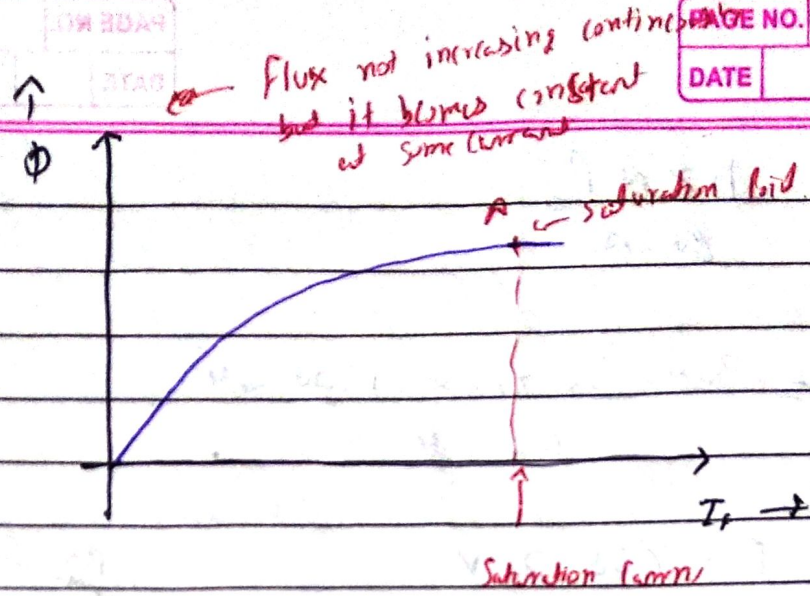
$$\rightarrow V_t \uparrow$$

$$\rightarrow I_f \uparrow = \frac{V_t \uparrow}{R_f}$$

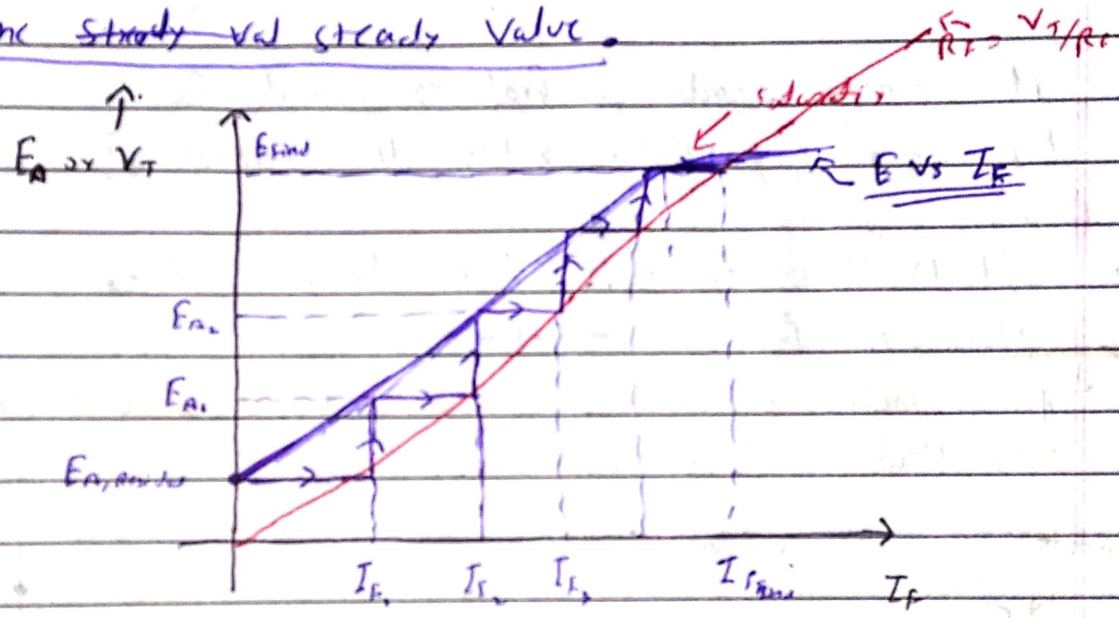
$$\rightarrow \phi \uparrow$$

← A loop is formed

$$\phi \propto I_f$$



After point A everything will become constant and I_F reach the steady val steady value.



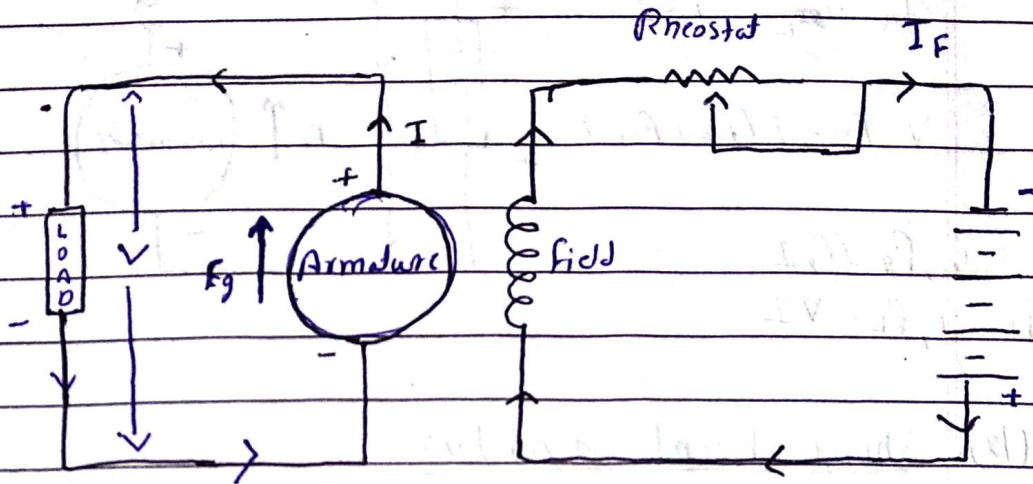
- Steps are just for explanation, E_A & I_F increases simultaneously until steady state condition.

⇒ Types of Generator's :-

(T) Separately - Excited DC Generator's →

Since the operation of a dc machine depends upon a fixed fixed Polarity of the Poles which does not vary with time, the field coils need energization from a dc source.

- An ac generator whose field winding is excited from an independent external dc source. The generator is called a Separately excited generator.



Armature current $\rightarrow$ Load current

$$I_a = I_L = I$$

Voltage across load $\rightarrow V = E_g - I R_a$ Armature Resistance

$\rightarrow$ voltage drop in armature

Power generated $\rightarrow P_g = E_g I$

Power delivered to external load $\rightarrow P_L = VI$

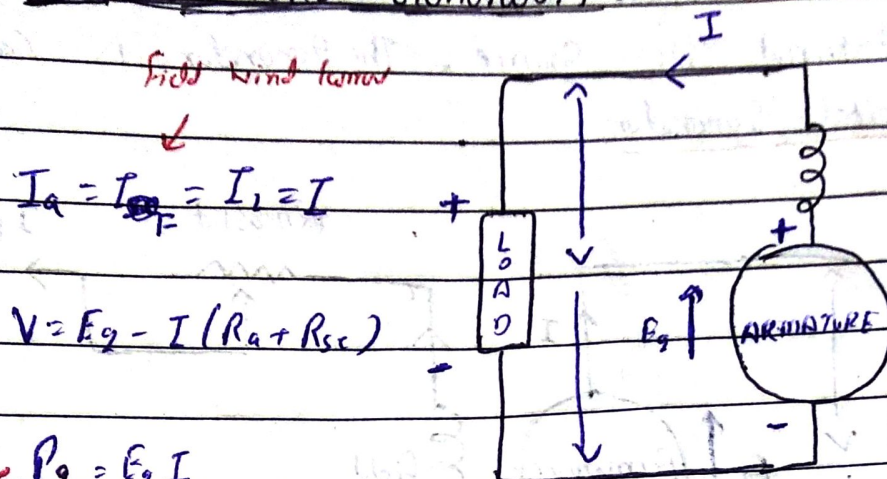
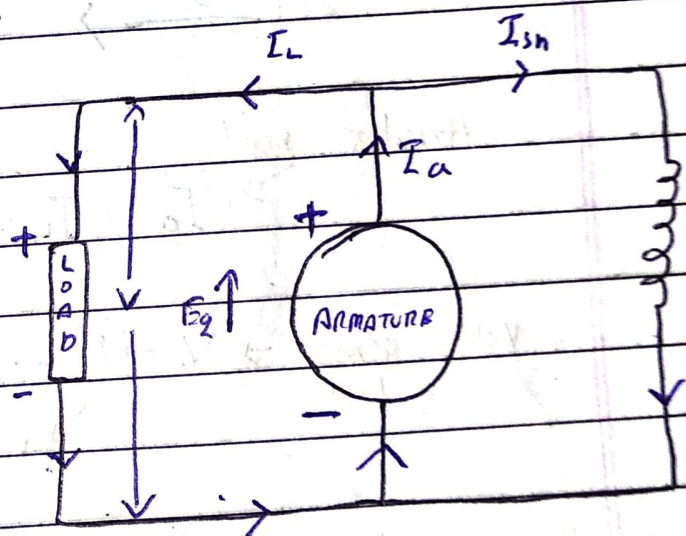
(II) Self-Excited DC Generator:-

- A dc generator whose field winding is excited by the current supplied by the generator itself, is called Self excited generator.

Series Wound
Generator

Shunt Wound
Generator

Compound Wound
Generator.

(a) Series Wound Generator's :-Power developed, $P_g = E_g I$ Power delivered, $P_L = VI$ (b) Shunt Wound Generator's :-Shunt Field winding $\rightarrow I_{sn} = \frac{V}{R_{sn}}$ Armature current $\rightarrow I_a = I_L + I_{sn}$ Terminal voltage $\rightarrow V = E_g - I_a R_a$ Power developed $\rightarrow P_g = E_g I_a$ Power delivered $\rightarrow P_L = VI_L$ 

⇒ Principle of Operation of DC Motor:-

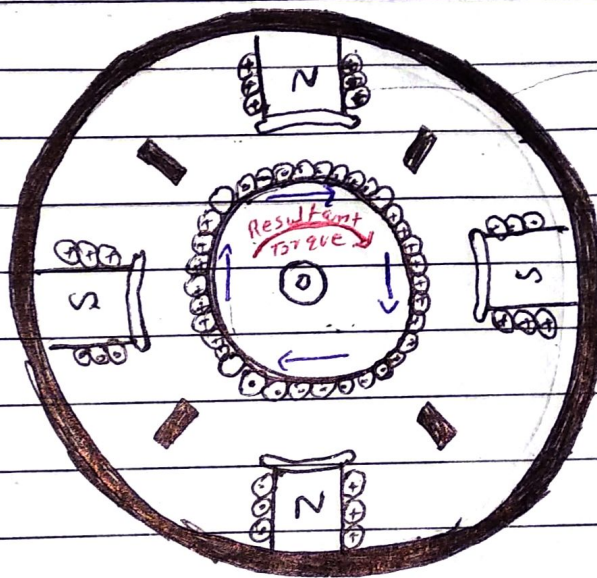
If a current carrying conductor is placed in a M.F, mechanical force is experienced on the conductor,

$$F = I \times B \times \text{length}$$

or,

$$I \times B \Rightarrow [F = I \times B]$$

← direction given by F.L.H Rule



POLE DC motor

Working :-

- When the DC motor is connected to DC supply, a direct current passes through the brushes and commutator to the armature winding, while it passes through the commutator it converts into AC so that the group of conductors under successive field poles carry currents in the individual opposite directions, also the direction of current in the individual conductors reverses as they pass away from the influence of one pole to that of next.

⇒ Back emf →

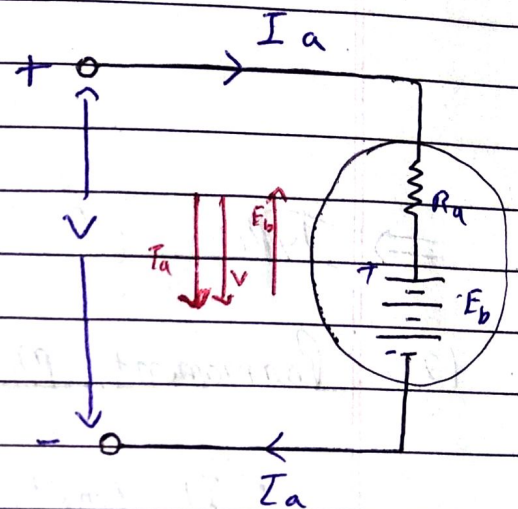
When the motor armature continues to rotate due to motor action, the armature conductor's cut the magnetic flux and therefore emf are induced in them.

The direction of this induced emf, known as back emf, is such that it opposes the applied voltage.

$$\text{Back emf, } \left[E_b = \frac{\Phi Z N}{60} \times \frac{P}{A} \right]$$

$$V = E_b + I_a R_a$$

armature current $\rightarrow I_a = \frac{V - E_b}{R_a}$



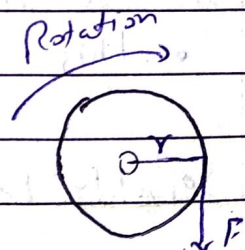
- The presence of E_b makes the dc motor a Self-Regulating machine, i.e. it makes the dc motor to draw as much armature current as is just sufficient to develop the required load torque.

⇒ Torque →

$$\tau = r \times F = r F \sin \theta$$

$$\tau = r F \quad (\because \theta = 90^\circ \text{ or } r \perp F)$$

$$\tau = r F \quad \text{--- (1)}$$



W.D Per revolution, $W = F \times 2\pi r$

W.D Per second, $W = F \times 2\pi r \times n$ $\rightarrow$ no. of revolution in 1-second $\rightarrow$ called Frequency

From ① $\rightarrow F_r = \tau \rightarrow 2\pi n \cdot (\omega) \rightarrow$ angular velocity

So,

$$W = \tau \omega \quad \text{J/s or watt}$$

~~Power developed, $P = T\omega = T \cdot 2\pi n$~~ Speed - Torque eqn

Power developed,

$$P = \tau \omega = \tau \times \frac{2\pi n}{60} \quad \begin{matrix} \rightarrow n \\ N \rightarrow \text{speed in r.p.m.} \end{matrix}$$

$$P = 0.105 NT \quad \text{watt}$$

$\Rightarrow$ Types of D.C Motor and their characteristics:

(I) Permanent Magnet motor:

- It consists of an armature and one or several permanent magnets encircling the armature.
- Field coils usually not required, However some of these motor's do have coils wound on the poles. these coils are intended only for recharging the magnets in the event that they loose their strength.
- this motor appears as only an armature.

(II) Separately Excited DC motor: $\rightarrow$ These motor's have field coils similar to those of a

shunt wound machine.

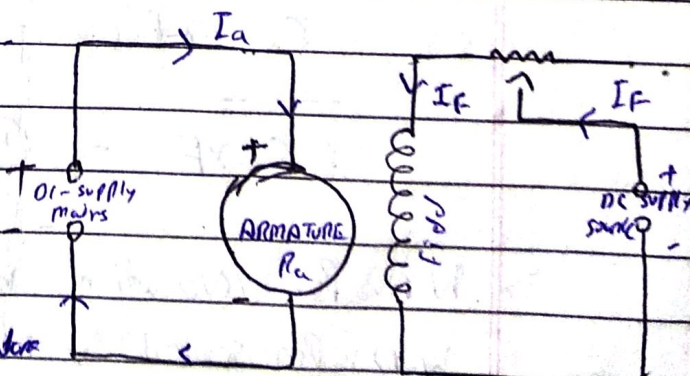
$$E_b = V - I_a R_a$$

$$P = IV$$

Mechanical power, $P_m =$ power input to armature -

power lost in armature

$$P_m = VI_a - I_a^2 R_a = I_a (V - I_a R_a) = IE_b$$



- Torque-armature current, Speed-armature current and Speed-torque characteristics of separately-excited dc motor are similar to those of dc shunt motor.

(III) Series Wound DC Motor → As the name implies, the field coils, consisting of few turns of thick wire, are connected in series with armature.

$$I_a = I_f = I \rightarrow \text{line current}$$

$$\text{Back emf developed, } E_b = V - I(R_a + R_f)$$

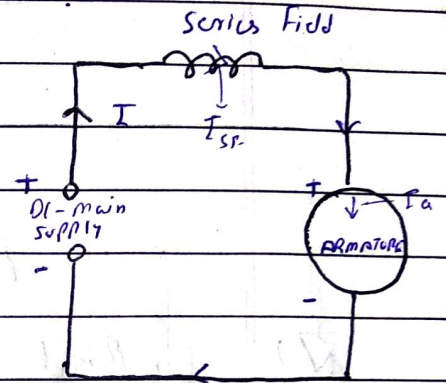
$$\text{Power drawn from main supply} = VI$$

$$\text{Mechanical Power, } P_m = \text{Power Input} - \text{Power losses in armature and field}$$

$$P_m = IV - I^2(R_a + R_f)$$

$$= I[V - I(R_a + R_f)]$$

$$P_m = I E_b$$

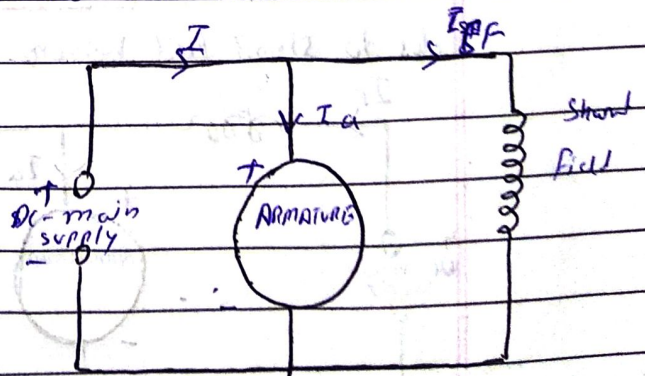


(IV) Shunt Wound DC motor:- The word "Shunt" means Parallel. These motor basically operate with the field coils connected in parallel with the armature.

$$I = I_a + I_{sf}$$

$$I_{sf} = \frac{V}{R_{sf}}$$

$$I_a = \frac{V}{R_a}$$



Back emf, $E_b = V - I_a R_a$
 Power drawn from main supply, $P = VI$

P_m = Power inpt $\rightarrow$ Power losses in armature and field.

$$P_m = IV - V I_{SF} - I_a^2 R_a$$

$$P_m = V(I - I_{SF}) - I_a^2 R_a$$

$$= V I_a - I_a^2 R_a$$

$$= I_a (V - I_a R_a)$$

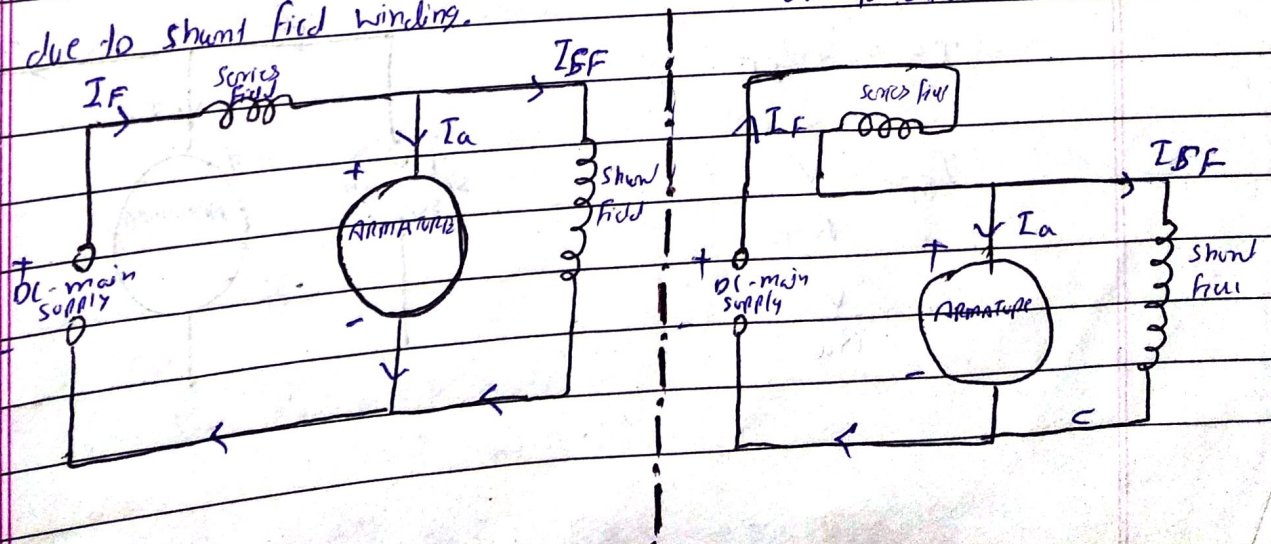
$$P_m = I_a E_b$$

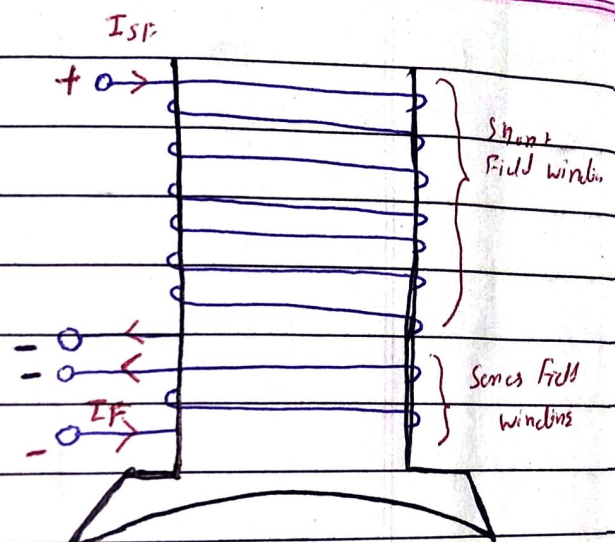
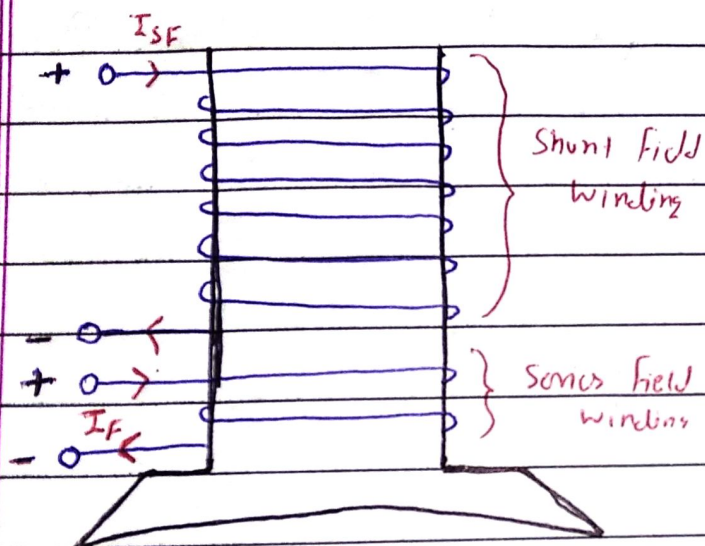
(✓) Compound wound DC motor:-

Cumulative Compound
wound motor

Differential Compound
wound motor

- The one in which the field windings are connected in such a way that the direction of flow of current is same in both the field windings.
- In motor of this type the flux due to series field winding strengthens the field due to shunt field winding.
- The one in which the field windings are connected in such a way that the direction of flow of current is opposite to each other in the two field windings.
- In this type of motor the flux due to series field winding weakens the field due to shunt field winding.





⇒ Speed Control of DC Motor:-

Speed Control means intentional change of the drive speed to a value required for performing the specific work process.

$$\text{Speed, } \left[N = \frac{k [V - I_a (R + R_a)]}{\phi} \right] \Rightarrow \left[N = \frac{k E_b}{\phi} \right]$$

$\phi \rightarrow \text{flux}$

⊙ Speed Regulation:- It is defined as the change in speed, when the load on the motor is reduced from full-load to zero-load and expressed in percentage of rated full-load speed.

$$\left[\text{Percentage Speed Regulation,} = \frac{N_0 - N_F}{N_F} \times 100 \right]$$

⊙ For a dc-series motor, after saturation,

flux is independent of field current I_f or armature current I_a and speed,

⇒ Apps of DC Generator:-

$$N \propto E_b$$

(1.) Separately-Excited DC Generator's:- These generator's require separate source of supply and are therefore, usually more expensive than self-excited. On this account these are generally used where self-excited generator's would be relatively unsatisfactory.

(2) Series Wound DC Generator's :- These generator's have very few application. They cannot be employed for power supply on account of their rising voltage characteristics. They can be employed as a constant current source by making use of the drooping portion of the voltage-current characteristics.

- Their best Application is for supplying field current for regenerative braking in dc locomotive.
- They are also used in series arc lighting.
- Another important application of these generator's is as (series booster) $\rightarrow$ it is a series wound generator, which is inserted in a circuit to add a certain voltage so that the excessive drop in the line is compensated.

(3) Shunt Wound DC Generator's

- They are used for light and power supply purpose.
- Also used for charging of batteries bcs they can be made to give constant output voltage.

(4) Comulative Compound Wound DC Generator's :-

- Generally used for lighting and power services bcs overcompounding compensates voltage drop in the distribution line and thus the voltage at the consumer terminal remains more or less constant.

(5) Differential Compound wound DC Generator's :-

- Useful application as an arc welding generator, where the generator is practically short circuited.

⇒ Application's of DC Motor:-

(1.) DC Series Motor:-

- The motor has a very high starting torque (up to 5-times of full-load torque), therefore it is used for drives requiring very high starting torque and where adjustable varying speed is satisfactory.
- Industrial uses are:- Hoists, cranes, trolley cars, conveyor's, elevator's, etc.

(2.) DC Shunt Motor:-

- Used essentially for constant speed application's requiring medium starting torque.
- May be used for adjustable speed not greater than 2:1 range.
- Industrial Uses are:- Lathes, centrifugal pumps, reciprocating pump's, fans, blowers, conveyor's, woodworking machines, machine tools, printing press, etc.

(3.) Compound Wound motor's:-

- ① Differential Compound Wound dc motor's are rarely used bcs of their poor torque characteristics and difficulties experienced during overloads and starting.
- ② Cumulative Compound Wound dc motor's - are used in driving machines which are subjected to sudden application's of heavy loads, such as occur in rolling mills, punching and shearing machines, lifts, mine hoists, etc.
 - This type of motor is also used ~~in~~ where a large starting torque is required but series motor cannot be conveniently used such as in cranes and elevator's.